

Foundation (Jalapeno)**Q1.** Which of the following is a relation?

- (A) A set of ordered pairs
- (B) A set of equations
- (C) A set of graphs
- (D) A set of functions
- (E) A set of tables

Q2. What is the vertical line test used for?

- (A) To find the domain of a function
- (B) To find the range of a function
- (C) To determine if a relation is a function
- (D) To evaluate a function
- (E) To graph a function

Q3. Given the function $f(x) = x^2$, what is the domain?

- (A) $(-\infty, \infty)$
- (B) $(-\infty, 0)$
- (C) $[0, \infty)$
- (D) $(-\infty, 0]$
- (E) $[0, \infty)$

Q4. Given the function $f(x) = \frac{1}{x}$, what is the range?

- (A) $(-\infty, 0)$
- (B) $(-\infty, \infty)$
- (C) $(0, \infty)$
- (D) $(-\infty, 0)$
- (E) $[0, \infty)$

Q5. What does the notation $f(2)$ represent?

- (A) The domain of the function f
- (B) The range of the function f
- (C) The value of the function f when $x = 2$
- (D) The equation of the function f
- (E) The graph of the function f

Q6. Given the table of values for a relation, which of the following is a function?

- (A) A table with two x -values that have the same y -value
- (B) A table with two y -values that have the same x -value
- (C) A table with one x -value that has two y -values
- (D) A table with one y -value that has two x -values
- (E) A table with one x -value that has one y -value

Q7. Which of the following is an example of a relation that is not a function?

- (A) A circle centered at the origin
- (B) A line with a slope of 2
- (C) A parabola that opens up
- (D) A horizontal line
- (E) A table with one x -value that has one y -value

Q8. Given the function $f(x) = x^2 - 4$, what is the value of $f(-2)$?

- (A) 0
- (B) -4
- (C) 4
- (D) -2
- (E) 2

Open Ended Questions (FRQ)**Q9.** Determine the domain and range of the function $f(x) = \frac{1}{x-2}$. Show all work and justify your answers.**Q10.** Evaluate the function $f(x) = x^2 - 3x + 2$ when $x = -1$. Show all work and justify your answer.**Chef's Tips**

- Read each question carefully
- Use proper notation and show all work
- Check your answers for reasonableness